Cayley graphs

Given a group Γ, and a symmetric set $S \subseteq \Gamma$ (i.e. $x \in S \Rightarrow x^{-1} \in S$), the Cayley graph $\mathrm{Cay}(\Gamma, S)$ is defined as follows:

$$V = \Gamma\,, \qquad E = \{\{x, y\} : xy^{-1} \in S\}$$

The random Cayley graph $\mathrm{Cay}_p(\Gamma)$ is defined, given $0 < p < 1$, as $\mathrm{Cay}(\Gamma, S)$ where each pair $\{s, s^{-1}\}$ ($s \neq e$) is included in S independently at random with probability p.

What is the typical behaviour of the clique number of $\mathrm{Cay}_p(\Gamma)$?

Proposition (Alon, Orlistky 1995):
There exists an absolute constant C such that, for any group Γ of order N, then

$$\omega(\mathrm{Cay}_{\frac{1}{2}}(\Gamma)) \leq C(\log N)^2$$

Pf Sketch (...)

Claim: For any $A \subseteq \Gamma$, there exists $B \subseteq A$ such that:

$$|B| = \Theta\big(\sqrt{|A|}\big)$$

$$|B^{-1}B| \geq \delta \cdot |B|^2$$

where δ is an absolute constant.

To prove the Claim:
We consider the multiplicative energy: For $X \subseteq \Gamma$,

$$E(X) := \#\{(x_1, x_2, x_3, x_4) \in X^4 : x_1^{-1}x_2 = x_3^{-1}x_4\}$$

For $q \in \Gamma$, let $r_X(q) = \#\{(x,y) \in X^2 : q = x^{-1}y\}$. Then

$$E(X) = \sum_{q \in \Gamma} (r_X(q))^2 \geq \frac{(\sum r_X(q))^2}{\#\{q : r_X(q) > 0\}} = \frac{|X|^4}{|X^{-1}X|}$$

If $E(X) = O(|X|^2)$, then

$$|X^{-1}X| \geq \frac{|X|^4}{E(X)} \geq \frac{|X|^4}{C|X|^2} = \delta |X|^2$$

Exercise: Show that for any $A \subseteq \Gamma$, there exists $X \subseteq A$ such that $|X| = \Theta\big(\sqrt{|A|}\big)$, and $E(X) = O(|X|^2)$.

Other results

Ben Green and Rob Morris showed that clique numbers of random Cayley graphs behave similar to random graphs $G(n,p)$, in the cyclic case $\Gamma = \mathbb{Z}/N\mathbb{Z}$:

Theorem (Green 2005) $\omega(\mathrm{Cay}_{\frac{1}{2}}(\mathbb{Z}/N\mathbb{Z})) = \Theta(\log N)$ w.h.p.
Theorem (Green, Morris 2015) $\omega(\mathrm{Cay}_{\frac{1}{2}}(\mathbb{Z}/p\mathbb{Z})) = (2+o(1))(\log p)$ w.h.p.

The main theorem we will prove generalizes Green's result for arbitrary groups.

Theorem (Conlon, Fox, Pham, Yepremyan)
For any group Γ, the following holds with high probability:

$$\omega(\mathrm{Cay}_p(\Gamma)) = O_p(\log N \log\log N) \qquad (\text{where } N = \Gamma)$$

Proof of the main theorem

Random entangled graphs

Given a coloring $c : E(K_n) \to [r]$, we consider the random graph $G_p(c)$ obtained by choosing to include each color class with probability p, independently. (i.e. take all edges of the chosen colors to be the edges of $G_p(c)$)

Note: The random Cayley graph is a random entangled graph, where we color each edge $\{x, y\} \in E(K_N)$ with the pair $\{xy^{-1}, yx^{-1}\}$.

Δ-bounded colorings

A coloring is Δ-bounded if every vertex is incident to no more than Δ many edges of the same color.

Note: The coloring associated to random Cayley graphs is 2-bounded: we can only have edges $\{x, y\}$, $\{x, z\}$ of the same color if $\{xy^{-1}, yx^{-1}\} = \{xz^{-1}, zx^{-1}\}$, which can only happen if $y = z$ or $y = xz^{-1}x$.

In light of this, in order to prove the main theorem, it is enough to show the following:

Theorem. If c is a Δ-bounded coloring of K_N, then w.h.p. we have

$$\omega(G_p(c)) = O_{p,\Delta}(\log N \log\log N)$$

To prove this theorem, we will use the first moment method together with the following counting lemma.

Lemma. There exists $C > 0$ such that, for any $K > 0$, and any Δ-bounded coloring $c : E(K_N) \to [r]$, the number M of subsets $A \subseteq [N]$ of size n, such that the number of colors in $K_N[A]$ is at most Kn, satisfies

$$M \;\le\; N^{C\Delta K \log N} \cdot (C\Delta K)^n$$